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DEPARTMENT OF PHOTOGRAMMETRY REMOTE SENSING OF ENVIRONMENT AND SPATIAL
ENGINEERING



PROM SHORT-TERM ACADEMIC EXCHANGE PROGRAMME

Research Exchange Report

Research cooperation in the field of remote sensing of environment

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1 Introduction

This report summarizes the activities undertaken during the short-term academic exchange at the Department of Photogrammetry Remote Sensing of Environment and Spatial Engineering in the AGH University of Kraków. The mobility took place from 14.09.2026 to 27.09.2026.

The main goal of this exchange was to detect and classify burned areas using different methods: Thresholding, Machine Learning, and Deep Learning. The workflow is divided into two stages. The first stage consists of basic image processing for satellite images and classical methods for burned area detection. The second stage involves detection methods using deep learning techniques, specifically the U-Net architecture. Finally, the results of both approaches will be compared.

2 The Area Of Interest

The area of interest is located in Los Gallardos, Almería, in the region of Andalucia, Spain. Specifically, the study focuses on the severe wildfire that started on July 9, 2026. This wildfire affected approximately 5200 hectares and required the evacuation of over 1400 people. Furthermore, the disaster resulted in 14 fatalities, and investigations suggest it may have been triggered by a fallen electric cable, highlighting the critical need for rapid detection and management of such events.

This event was the third-deadliest wildfire in Spain's history. Due to its massive scale and impact, widespread spatial analysis was generated, with several scientific institutions employing remote sensing tools to understand its evolution.

3 Methodology and Results

The methodology was structured to compare traditional remote sensing techniques, classical machine learning, and modern artificial intelligence approaches.

3.1 Stage 1: Basic Image Processing and Spectral Indices

The first stage focused on classical methods for burned area detection.

- Jupyter notebooks were utilized for warm-up exercises involving libraries such as numpy, scikit-image, and rasterIO.
- Sentinel-2 satellite data was collected for the periods before and after the fire event.
- Spectral indices, specifically the Normalized Burn Ratio (NBR) and delta NBR (dNBR), were calculated to exploit the spectral signatures of the land surface.
- Classical thresholding methods were applied to these indices to binarize and identify the burned areas, followed by morphological operations to remove noise and refine the structural elements of the binary masks.

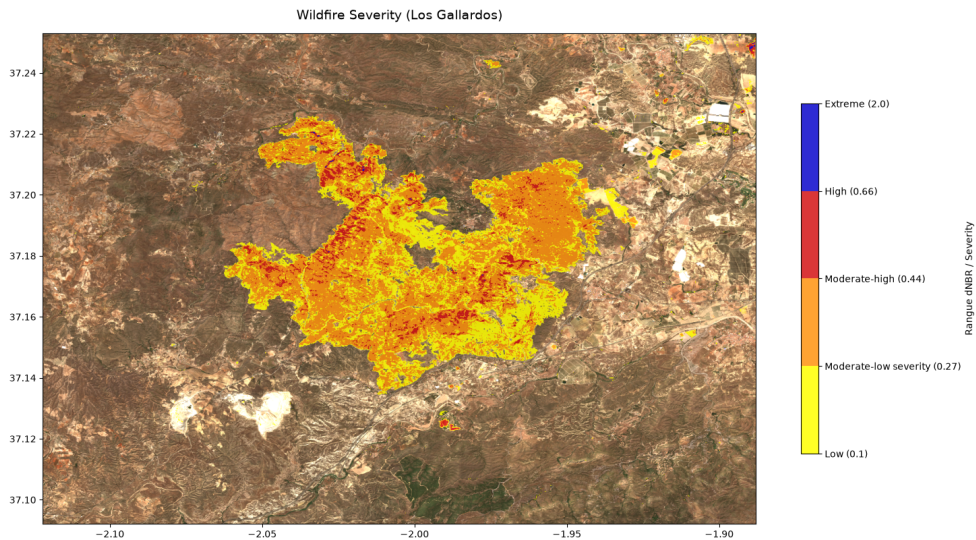


Figure 1: Fire severity map of Los Gallardos calculated using the dNBR spectral index.

3.2 Stage 2: Machine Learning Approaches

As an intermediate step, supervised machine learning classification methods were considered to group pixels based on their spectral similarity.

- Algorithms such as Random Forest, Support Vector Machines (SVM), and XGBoost were explored to learn how to assign pixels to predefined classes using labeled training data.

3.3 Stage 3: Deep Learning (U-Net)

The final stage implemented advanced deep learning techniques to detect the same burned areas, focusing on the U-Net architecture.

- The U-Net model, which is designed for pixel-wise image segmentation, was implemented to extract hierarchical features.
- This model uses an encoder-decoder architecture with skip connections that help preserve fine-grained spatial information lost during downsampling, capturing the complex spatial patterns of wildfire spread.
- The network works well with limited annotated data when augmentation is applied, making it highly effective for satellite imagery.

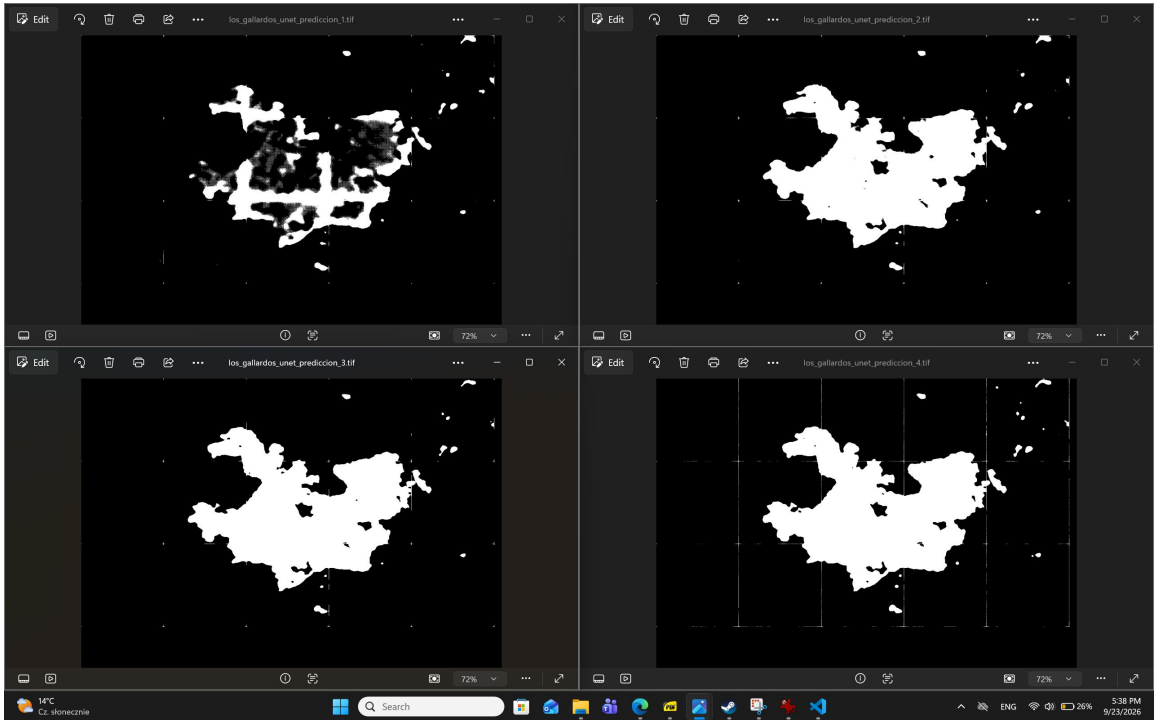


Figure 2: Individual image patch predictions generated by the U-Net model during the inference phase.

3.4 Comparison and Evaluation Metrics

The final phase consists of comparing the burned area masks generated by the basic image processing and machine learning techniques against the predictions made by the U-Net model. To quantitatively assess the performance, four key evaluation metrics were used:

- **Intersection over Union (IoU):** Quantifies the overlap between predicted and actual burned areas.
- **Precision:** Measures the proportion of correctly predicted burned pixels, indicating the minimization of false positives.
- **Recall:** Evaluates the model's ability to capture all true burned regions (minimizing false negatives).
- **F1-score:** Provides a balanced harmonic mean of Precision and Recall for overall classification accuracy.

3.5 Results and Visualization

The quantitative and qualitative results demonstrated that the U-Net architecture significantly outperformed classical thresholding approaches. While dNBR thresholding struggled with false positives in areas with sparse vegetation or shadows, U-Net successfully generalized the complex spatial patterns of the burn scar.

Quantitatively, the U-Net model achieved an outstanding Recall of 0.9949, indicating that it successfully identified 99.49% of the actual burned pixels. The Precision was 0.6252, which, combined with the Recall, resulted in a strong and balanced F1-Score of 0.7678. Furthermore, the model demonstrated a solid spatial overlap with the traditional thresholding mask, recording an Intersection over Union (IoU) of 0.6232.

Following the prediction phase on the 256x256 image patches, a mosaicking process was applied to reconstruct the full georeferenced map. The final outputs were successfully exported and visualized using QGIS, visually confirming the high precision of the neural network in delineating the 5200-hectare affected zone.



Figure 3: Final binary mask of the burned area reconstructed from the U-Net semantic segmentation.

4 Improvement Factors and Advantages

The implementation of deep learning models, specifically the U-Net architecture, introduces several significant improvement factors compared to traditional thresholding methods. The main advantages of neural network-based segmentation include:

- **High Accuracy:** The model is highly accurate because it automatically learns complex features and patterns.
- **Versatility:** U-net model works on any image type, being useful for natural images, medical scans, and satellite imagery, although its performance should be further evaluated in other environments.
- **Handling Complex Objects:** The architecture is capable of handling complex objects but an overlapping strategy should be applied during the image reconstruction phase to reduce visible seam lines between merged segments.
- **Automated Feature Extraction:** Deep learning automates the feature extraction process, eliminating the need for manual feature engineering, however, additional filtering techniques should be applied to reduce noise in the final output.
- **Spatial Precision:** Through the use of skip connections, U-Net combines global context with fine spatial details, which is ideal for segmenting the complex and irregular spatial patterns left by a wildfire.
- **Data Efficiency:** The network can be trained end-to-end and works well even with a limited amount of annotated data when data augmentation techniques are applied.
- **Processing Speed:** The model is highly efficient during the inference phase, taking less than a second to segment a 512x512 pixel image on a recent GPU.
- **Patch-based Processing:** It efficiently processes large satellite maps by dividing the data into smaller patches (e.g., 256x256 pixels) to learn and classify each pixel individually.

While this project serves as a strong basis, there are still significant opportunities for improvement, such as refining the script, testing the model in different geographical areas, and applying post-processing filters to reduce noise.

```
· Loading maps for evaluation...
  Loading metrics...
  -----
  U-NET vs CLASSIC dNBR EVALUATION RESULTS
  -----
  Intersection over Union (IoU): 0.6232
  F1-Score:                      0.7678
  Precision:                     0.6252
  Recall:                        0.9949
  -----
```

Figure 4: Results of metrics.

5 Conclusions

The short-term exchange facilitated a comprehensive and comparative analysis of the Los Gallardos wildfire, demonstrating the superiority of modern artificial intelligence over traditional remote sensing techniques. The U-Net architecture outperformed classical dNBR thresholding by automating feature extraction and eliminating the need for manual threshold adjustments. The model’s success is largely driven by its encoder-decoder structure and skip connections, which effectively preserve fine-grained spatial details lost during downsampling, allowing it to capture the complex spatial patterns of wildfire spread.

With an Recall of 0.9949 and a robust F1-Score of 0.7678, the deep learning approach proved reliable in delineating the affected area. Ultimately, transitioning from static mathematical formulas to pixel-wise semantic segmentation provides a highly accurate and scalable solution for rapid environmental monitoring. While further refinement in overlapping strategies and noise reduction is needed, this framework sets a strong foundation for future automated disaster management across diverse geographical landscapes, the flexibility of this approach allows for continuous model optimization and the integration of additional variables to adapt to evolving research needs.

Acknowledgments

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